

CHAPTER 1

INTRODUCTION

The World Health Organization (WHO) (2016) stated that hundreds of million of people are affected by neurological disorders wherein Alzheimer's Disease (AD) is the most common cause of dementia and contributes 60-70% of the cases [1]. They also estimated that 1 in 160 children has an Autism Spectrum Disorder (ASD), and cases has been increasing globally ever since [2].

Alzheimer's Disease is characterized by deterioration and loss of memory and thinking skills due to permanent damage to the brain cells because the brain cortex particularly the hippocampus shrinks [5]. Autism Spectrum Disorder is characterized by communication deficiency, impaired social and emotional reciprocity, and irregular patterns of behavior, which mainly affects social interaction behavior of the person [6].

It is projected that about 66% increase in Alzheimer and other dementia cases will be observed from 2005 to 2030. Currently, patients' neurological disorders are diagnosed by evaluating neurological history and conducting examinations which are used to assess how the nervous system has been affected [3]. Today, Diagnosis of neurological disorders are the most challenging task due to the complexity of the nervous system. Methods to diagnose those disorder are done with the help of using neuroimaging technologies, such as EEG, MR Imaging, CT Scan, and PET [4].

Traditionally, diagnosis of Attention Deficit Hyperactivity Disorder (ADHD) was unreliable because of its basis in behavioral symptoms. Recently, Machine learning algorithms on neuroimages are being explored for biomarkers in ADHD [8]. Some studies have used deep learning to identify and classify Computed Tomography (CT) images, such as pneumonia,

tuberculosis, and brain tumors [9]. These technologies and methods are used to determine whether the patient being diagnosed is grouped together with healthy group or not.

From recent studies, identification and classification using deep learning increases the speed and accuracy of object detection. Identification of Autism and Alzheimer may see an increase in speed and accuracy of diagnosis in neurological disorders by using deep learning methods through feature extractions and their classification. (NOTE: THIS PART SHOULD BE DISCUSS FURTHER TO ESTABLISH THE OPPORTUNITY)

GAP / OPPORTUNITY

Brain MRI contains data which contains features which will be used to classify the differences between the patients. Deep learning may be used to extract features and segment the brain data, and later be used to diagnose the neurological condition of each MR data. Convolutional neural networks (CNN) is a deep learning model and an architecture in artificial neural networks. It's used in feature extraction, recognition, classification and detection [4]. There are challenges when using deep learning to neuroimaging such as data which are high resolution structural MRI or resting-state functional MRI wherein these data contains 0.1 to 1 million voxels for each subject data. In addition, usage of deep learning requires thousands of labelled data for better accuracies in classification [7].

The experiment starts with feature extraction, what kind of deep learning method should we use to precisely extract the data without losing too much? Considering the dimension of MR image dataset, there are many ways to extract features and segment like usage of 2D and 3D based network models. As MR image also contains skull, will removing or retaining it cause any changes in classification accuracies? Once extracted, what method should be used to classify each feature

and different them from each other? The problems situated in this research involves the inquiry on what methods and neural network models to use.

This study begins by addressing the issues from extracting the brain features from the MRI dataset then will be classified by conducting through a neural network. This study addresses the difference of brain structures and functions of a normal patient to a patient with either autism spectrum disorder or Alzheimer's disease. More particularly, it identifies the classification of the patients' MRI data based from the features extracted.

Problem Statement

This study's objective is to identify, classifying, and compare the ASD and AD group from the Normal Control (NC) group, which consists of MR images that contains healthy brain. The identification is grounded based on the differences of neurological anatomy and functionality observed from the MR images by the proposed neural networks.

Specifically, it aims to answer the following questions:

- 1) What extraction method is best suited to fully extract the brain features from the MR images with as much low loss as possible?
- 2) What model should be used to accurately classify the extracted brain features from each other?
- 3) What is the neurological functionality and anatomical the differences between the AD, ASD, and NC groups?

Objectives

(Note: write here equivalent objective statement of the research question)

Significance of the Study

The study focuses on identifying the difference between AD, ASD, and NC groups. Through usage of neural network to extract features and classify and identify, diagnosis of AD and ASD will be faster and the observations or maintenances performed by neurologist will be easier to achieve. By differentiating and comparing the differences between these groups, the results will be beneficial to the following:

- 1) Radiologist– They will greatly benefit from this study as they specialize in diagnosing and treating diseases using medical imaging such as MRI, X-Ray, Ct Scan, etc. This study will automate the identification and classification of neurological anomaly from the MR images.
- 2) Neurologist – The result of the studies will help them diagnose any neurological disorders and/or disease observed from the MR neuroimages of their patients.
- 3) Patients – As the task of neurologist is more automated and observed in more depths, their diagnosis will arrive sooner and more precisely thus resulting in early detection of diseases and disorders.

1.1 Scope and Limitations of the Study

The study focuses on differentiating the normal control, Alzheimer’s disease, and autism spectrum disorder group based from the MR images. The method to conduct the experiment will be using neural networks for feature extraction and classification.

The scope of the study contains the methods used for feature extraction of MR images and the neural network or model used to classify the sample data. Neural networks specifically convolutional neural networks will be the proposed method to extract the feature and identify and classify the data.

The dataset will be provided by a Filipino local government unit and scanned from Filipino patients from varying ages and sex. This study is expected to be finished within a span of 6 months. Initially, to develop and test the methods this study will be using publicly available datasets for AD and ASD. For the final experiment, the MR image dataset will be provided by a local government unit.

Techniques will involve performing feature extraction on the MR images, to segment and extract the required brain slices such as gray matter, white matter, frequency, etc. The extracted features will be classified and analyzed by the neural network to identify and compare between the ASD, AD and NC dataset.

This research will be limited to only classification and identification of only ASD and AD, no other neurological disorders or diseases will be included. In terms of identification, this study will be following current medical knowledge in AD and ASD and will not be containing medical research on the domain of neurology in general. Though this research tackles the domain of method and technologies in neuroimaging, it will not be including other medical imaging format other than MR imaging. Therefore, methods on brain features and the diagnosis of the neurological disorder through neural networks will be the focus of this study.

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